ESTR3114: Numerical Optimization

Course Project Proposal

Project Title: Gradient Conflict in Joint Token Prediction and Sentiment Classification

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Abstract

Standard language model training minimizes next-token cross-entropy. When an auxiliary objective is added—here, text-level sentiment classification—the two losses share the same Transformer parameters and may produce conflicting gradients. We study this setting as a multi-objective, non-convex optimization problem. A pretrained GPT-2 Small model will be fine-tuned on the three-class TweetEval sentiment dataset with (i) next-token prediction on each text and (ii) positive / neutral / negative classification using the final non-padding token representation. We compare standard AdamW optimization of the scalarized loss

$$\mathcal{L} = \mathcal{L}_{\text{tok}} + \lambda \mathcal{L}_{\text{sent}}$$

against Projected Conflicting Gradients (PCGrad). When the shared-parameter gradients have a negative inner product, PCGrad projects out their opposing components before the AdamW update. The goal is to measure how often and where gradient conflict occurs, whether it correlates with slower or less stable joint training, and whether PCGrad improves the token–sentiment trade-off relative to carefully tuned linear scalarization.

Planned Activities

- **Readings:** Yu et al., *Gradient Surgery for Multi-Task Learning* (PCGrad); Sener & Koltun on multi-objective multi-task learning; optional GradNorm for adaptive loss balancing.
- **Implementation:** Add a lightweight sentiment head to GPT-2 Small and implement separate task-gradient extraction and PCGrad on the shared backbone. The language-model and sentiment-specific heads receive their ordinary task gradients.
- **Experiments:** Compare (1) token-only, (2) sentiment-only, (3) fixed-weight joint AdamW, and (4) PCGrad under matched initialization, batches, learning-rate schedule, and step budget. Select λ in a short validation pilot, then run the main comparison with three random seeds.
- **Analysis:** Track layer-wise gradient cosine and conflict frequency, task gradient-norm ratios, token loss / perplexity, sentiment accuracy and macro-F1, training stability, and wall-clock overhead. Plot the empirical validation trade-off between the two objectives.
- **Success criteria:** Determine whether meaningful gradient conflict appears, quantify its layer-wise pattern, and evaluate whether PCGrad reduces measured conflict or improves the stability or empirical trade-off relative to linear scalarization.
- **Deliverables:** Reproducible code, gradient-conflict and convergence figures, discussion of positive or negative findings, and a final report linking the results to multi-objective first-order optimization.